

Project 4:

Predicting Spotify Charts

with Decision Tree Models

Presented by: Raven Otero-Symphony

Agenda

1

Project Goals

Clearly define the "success metric" and thoroughly clean the data before fitting any decision tree models.

2

Methodology

Incorrect data types by batch, missing data, visualizing variable relationships, correlation matrices, etc.

3

Model Results

What worked, what went wrong, how the model can be improved, key assumptions and future iterations.

4

Future Development

Key observations, revisions, adjusting model approach and alternative methods in data cleaning.

Project Overview

Objective: Minimize the Root Mean Squared Error (RMSE) in a Decision Tree Model to predict the popularity of a new song.

1,556

rows

22

columns

0

nulls?

popularity

true “success”
metric?

Data Source: AWS Spotify Streams 2020-2021

Only 2 data types?

Methodology

- **days_charted** = New success metric: total days charted * number of times charted
- Removed whitespace to accurately read-in **null** values
- Dropped **genre** (nunique = 395) and **song id**

1,485

rows

31

columns

7

Int64

2

datetime

7

float64

2

strings

regex

one-hot encoding

drop null rows

parse dates

"Highest Charting Position" is type: * int64 *

"Number of Times Charted" is type: * int64 *

"Streams" is type: * object *

"Artist Followers" is type: * object *

"Popularity" is type: * object *

"Duration (ms)" is type: * object *

"Tempo" is type: * object *

	Highest Charting Position	Number of Times Charted	Streams	Artist Followers	Popularity	Duration (ms)	Tempo
Index							
1	1	8	48,633,449	3377762	100	211560	134.002
2	2	3	47,248,719	2230022	99	141806	169.928
3	1	11	40,162,559	6266514	99	178147	166.928
4	3	5	37,799,456	83293380	98	231041	126.026
5	5	1	33,948,454	5473565	96	212000	149.995
...	...	...	...	...	...	...	...
1552	195	1	4,630,675	27167675	79	209320	116.073
1553	196	1	4,623,030	15019109	66	181930	152.37
1554	197	1	4,620,876	22698747	81	217307	104.988
1555	198	1	4,607,385	208630	60	152784	154.064
1556	199	1	4,595,450	42227614	70	221307	205.272

1556 rows x 7 columns

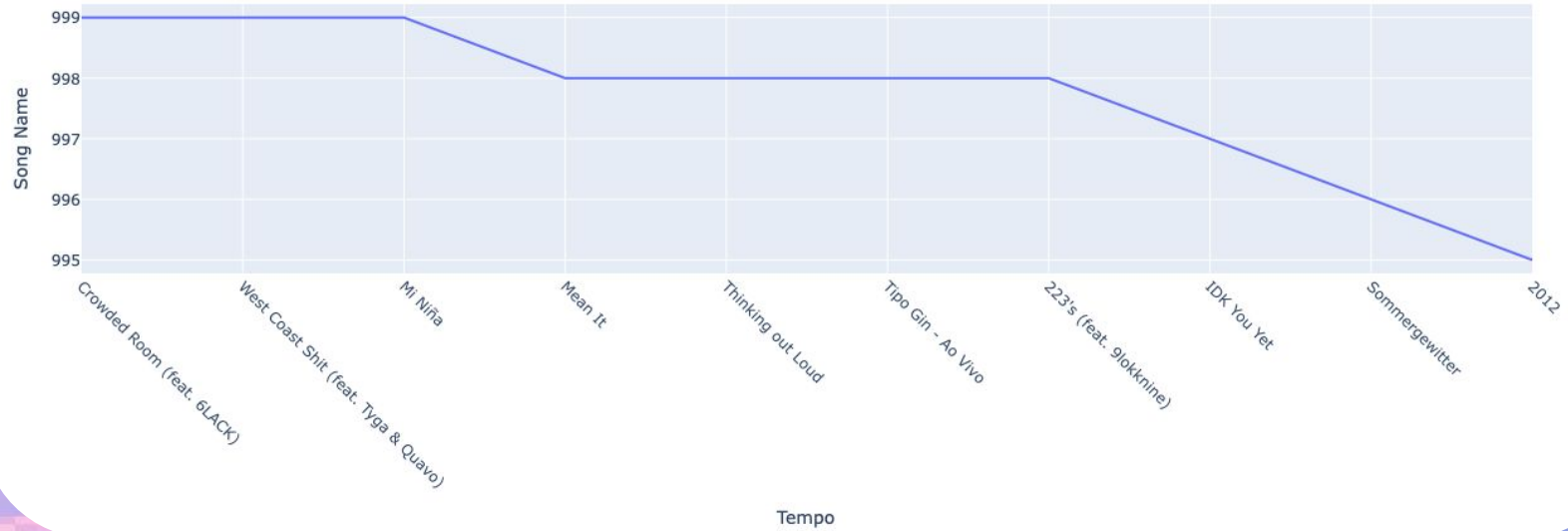
Int64 Challenge Snapshot

	streams	tempo	highest_position	times_charted	followers	popularity	duration_ms	danceability	energy	loudness	...	B	C	C#/Db	D
Index															
1	48633449	134	1	8	3377762	100	211560	0.714	0.800	-4.808	...	True	False	False	False
2	47248719	169	2	3	2230022	99	141806	0.591	0.764	-5.484	...	False	False	True	False
3	40162559	166	1	11	6266514	99	178147	0.563	0.664	-5.044	...	False	False	False	False
4	37799456	126	3	5	83293380	98	231041	0.808	0.897	-3.712	...	True	False	False	False
5	33948454	149	5	1	5473565	96	212000	0.736	0.704	-7.409	...	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1552	4630675	116	195	1	27167675	79	209320	0.762	0.700	-6.021	...	False	False	False	False
1553	4623030	152	196	1	15019109	66	181930	0.528	0.870	-3.123	...	True	False	False	False
1554	4620876	104	197	1	22698747	81	217307	0.765	0.523	-4.333	...	False	False	False	True
1555	4607385	154	198	1	208630	60	152784	0.832	0.550	-7.026	...	False	False	False	False
1556	4595450	205	199	1	42227614	70	221307	0.448	0.603	-7.176	...	False	False	False	False

1556 rows x 30 columns

Final DataFrame

Top 10 Songs with Impossible BPM

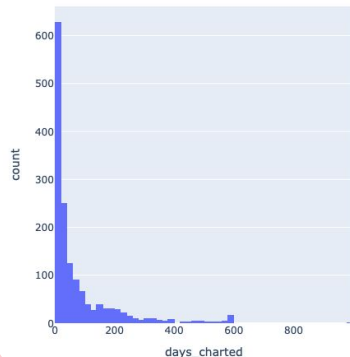
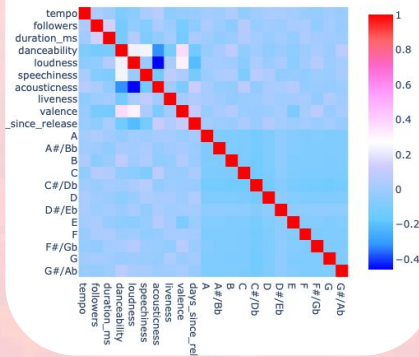


“Tempo” Observation: Skewed Data

Initial Model: Results

The “benchmark” model performed poorly due to a highly right-skewed target variable, outliers in the feature set, and duplicate values.

Selected Features: Correlation Matrix



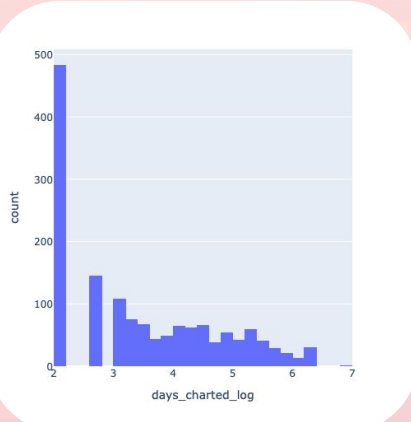
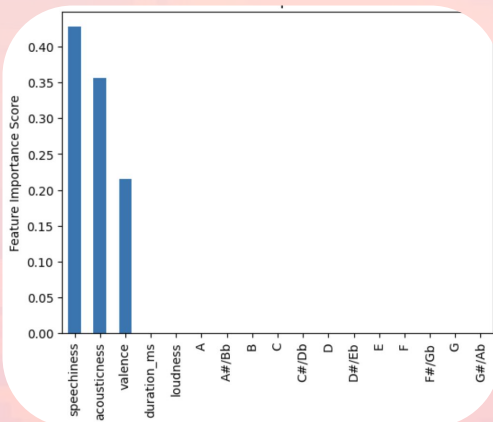
CV RMSE: 113,130 days

500 loops, **3** max_depth,
80/20 train/test split

Confirmed “popularity”
isn’t the most reliable
success metric

Final Model: Results

Model improved after log transformation of target variable, dropping columns with 60-70% nulls, and checking optimal max depth.



CV RMSE: 1,280 days

500 loops, **2** max_depth,
80/20 train/test split

Top features suggest
reliable internal metrics
for company goals

Future Development

- Refactor code
- Feature transformation & optimization
- Modeling in data type “batches” (e.g., internal metrics v. mix)
- Clean and group by genre (e.g., pop, rock, hip hop, etc.)
- Different correction methods for skew beyond log-transform

```
['indie rock italiano', 'italian pop'], ['australian hip hop',  
 'pop'], ['pop', 'uk pop'], ['lgbtq+ hip hop', 'pop rap'],  
 ['dance pop', 'pop'], ['puerto rican pop', 'trap latino'],  
 ['latin', 'reggaeton', 'trap latino'],  
 ['dance pop', 'pop', 'uk pop'],  
 ['latin', 'reggaeton', 'reggaeton colombiano'],  
 ['k-pop', 'k-pop boy group'],  
 ['canadian pop', 'pop', 'post-teen pop'],  
 ['canadian contemporary r&b', 'canadian pop', 'pop'],  
 ['reggaeton', 'trap latino'],  
 ['gauze pop', 'indietronica', 'shiver pop'],  
 ['latin', 'latin hip hop', 'reggaeton', 'trap latino'],  
 ['dfw rap', 'melodic rap', 'rap'],  
 ['dance pop', 'house', 'pop dance', 'tropical house', 'uk dance'],  
 ['dance pop', 'pop', 'post-teen pop'],  
 ['dance pop', 'electropop', 'pop', 'post-teen pop'],  
 ['colombian pop', 'latin', 'reggaeton', 'reggaeton colombiano'],  
 ['pop', 'post-teen pop'], ' ', ['chicago rap'],  
 ['latin', 'panamanian pop', 'reggaeton', 'trap latino'],  
 ['latin pop', 'pop argentino'],  
 ['big room', 'brostep', 'dance pop', 'dutch edm', 'edm', 'electro house',  
 'house', 'trance', 'tropical house'],  
 ['dance pop', 'edm', 'electro house', 'pop', 'pop dance', 'tropical h  
 ['cali rap'], ['trap argentino'], [''],  
 ['argentine hip hop', 'pop venezolano', 'trap argentino', 'trap latin  
 ['electropop', 'pop'], ['house', 'pop dance', 'uk dance'],  
 ['brostep', 'edm', 'pop', 'pop dance', 'progressive electro house'],  
 ['dreamo', 'indie surf', 'surf punk', 'vegas indie'],  
 ['pop', 'pop rock'],  
 ['dance pop', 'piano rock', 'pop', 'pop rock'],  
 ['houston rap', 'pop', 'pop rap', 'trap queen'],  
 ['german dance', 'german techno', 'german trance', 'trance'],  
 'pop', 'r&b']
```

[[View Github](#)]

Thank you!